Employee Ratings Application

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Overview

The Employee Ratings Application is a Python console program that manages employee performance review data. It allows users to view, enter, and save employee ratings using a layered architecture that separates data, processing, and presentation concerns.

Data is persisted to a JSON file (`EmployeeRatings.json`) which is read automatically on startup and written on demand. All employee records are stored in memory as a list of `Employee` objects during the session.

Features

- Load employee ratings automatically on startup from a JSON file
- Display all employee ratings with descriptive labels:
  - 5 = Leading,
  - 4 = Strong,
  - 3 = Solid,
  - 2 = Building,

    - 1 = Not Meeting Expectations
- Enter new employee ratings with full input validation
- Menu input loop forces a valid choice — invalid entries prompt a re-try rather than accepting bad input
- Save all ratings back to the JSON file
- Structured error handling throughout all layers
- Comprehensive unit test suite covering all modules (55 tests)

File Structure

```
project/
├── main.py                      # Application entry point
├── data_classes.py              # Person, Employee classes + constants
├── processing_classes.py        # FileProcessor class (JSON I/O)
├── presentation_classes.py      # IO class (user input/output)
├── EmployeeRatings.json         # Persistent data store
├── test_data_classes.py         # Unit tests for data classes
├── test_processing_classes.py   # Unit tests for FileProcessor
└── test_presentation_classes.py # Unit tests for IO
```

Architecture

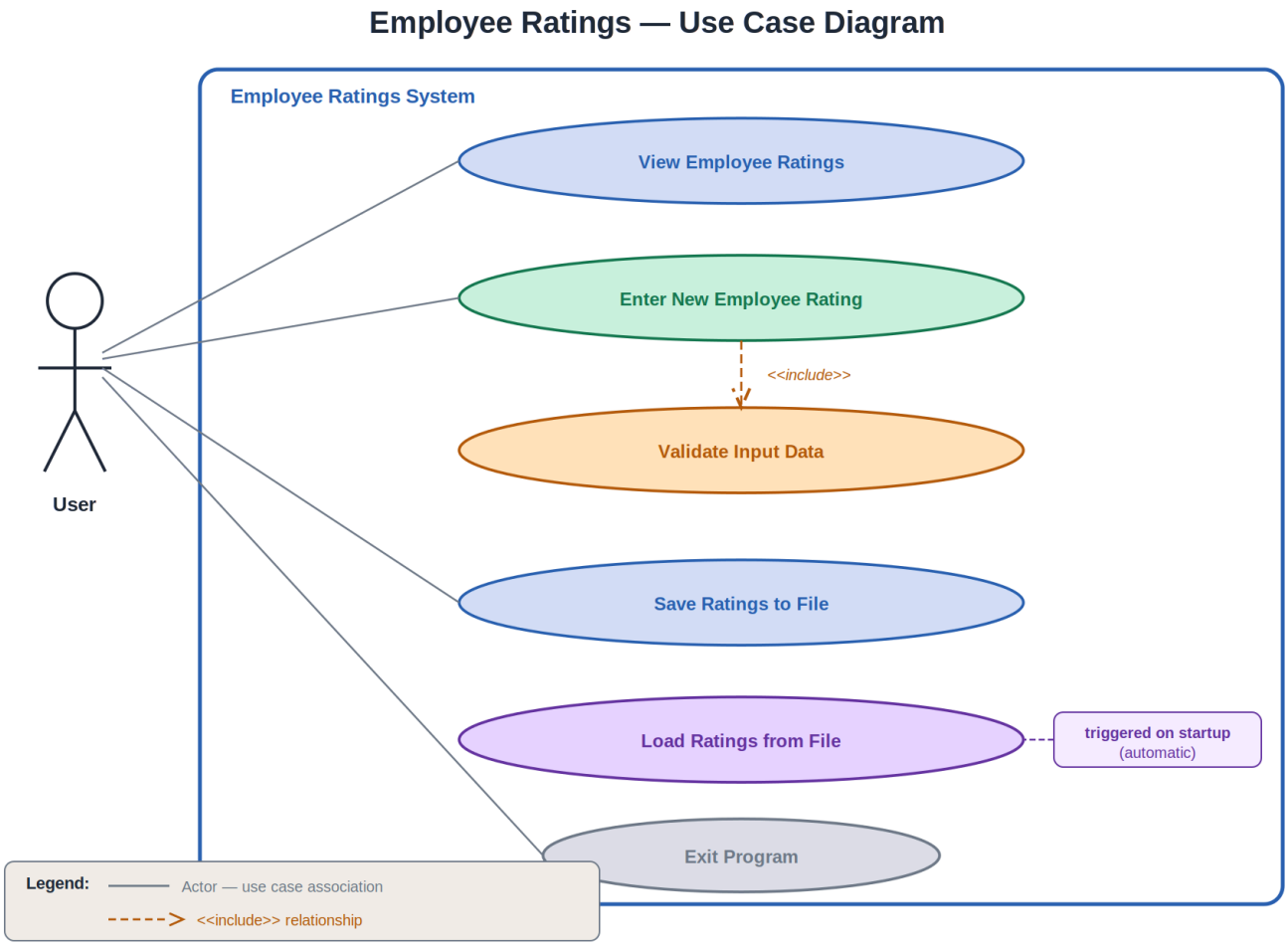
The application follows a three-layer architecture with a dedicated entry point:

Layer	Module	Responsibility
Data	`data_classes.py`	Classes, constants, variables
Processing	`processing_classes.py`	JSON file read and write operations
Presentation	`presentation_classes.py`	User input, output, error display
Entry Point	`main.py`	Orchestrates all layers

`main.py` imports all three supporting modules. `presentation_classes.py` and `processing_classes.py` both import `data_classes.py` for type references. There is no circular dependency between modules.

The `if __name__ == "__main__":` guard in `main.py` ensures the application loop does not execute when the module is imported by the test files.

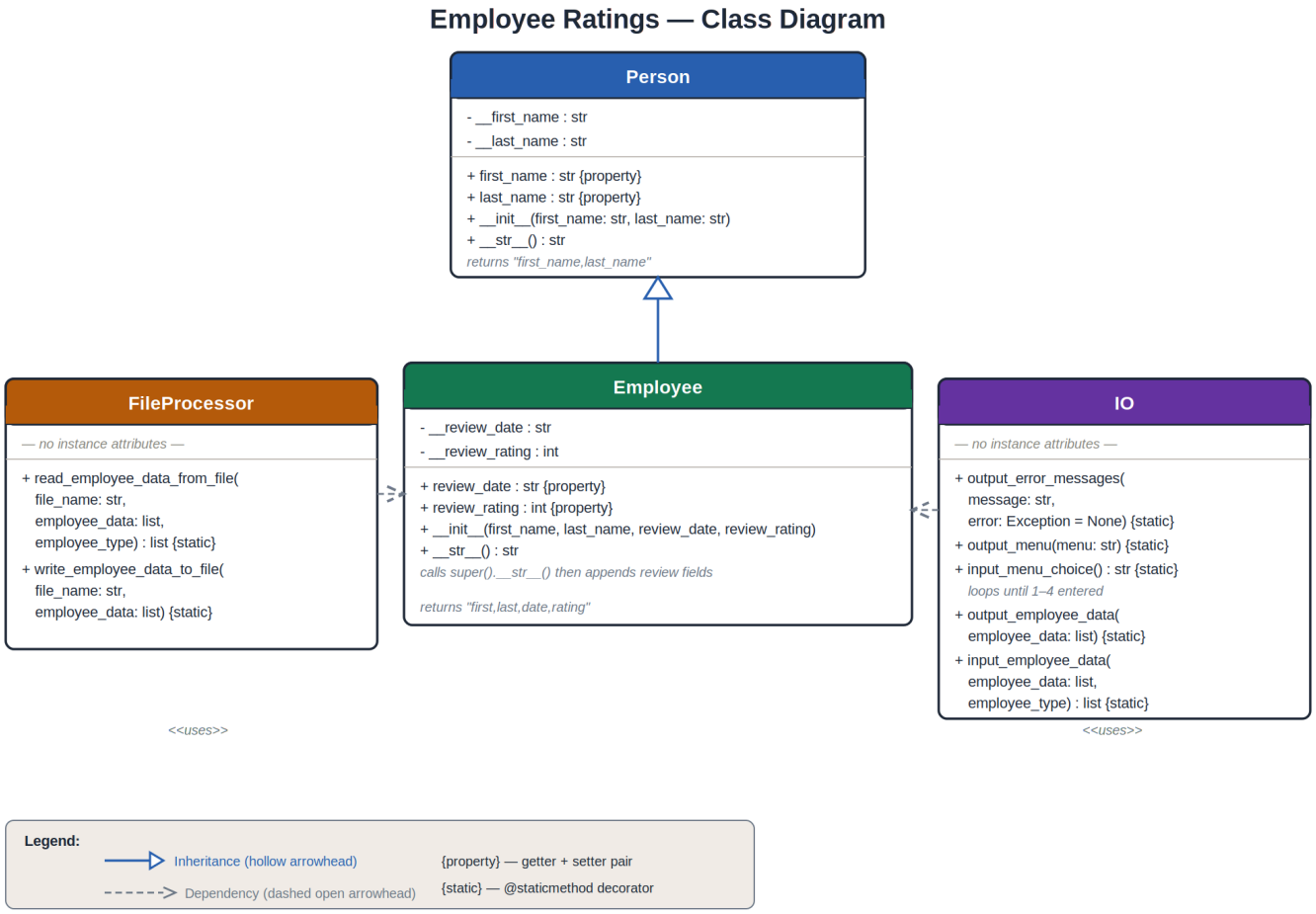
Use Case Diagram



The use case diagram shows how the User interacts with the system:

- **View Employee Ratings** — displays all current records
- **Enter New Employee Rating** — prompts for name, date, and rating; includes input validation
- **Save Ratings to File** — persists the in-memory list to JSON
- **Load Ratings from File** — runs automatically on startup
- **Exit Program** — terminates the application

Class Diagram

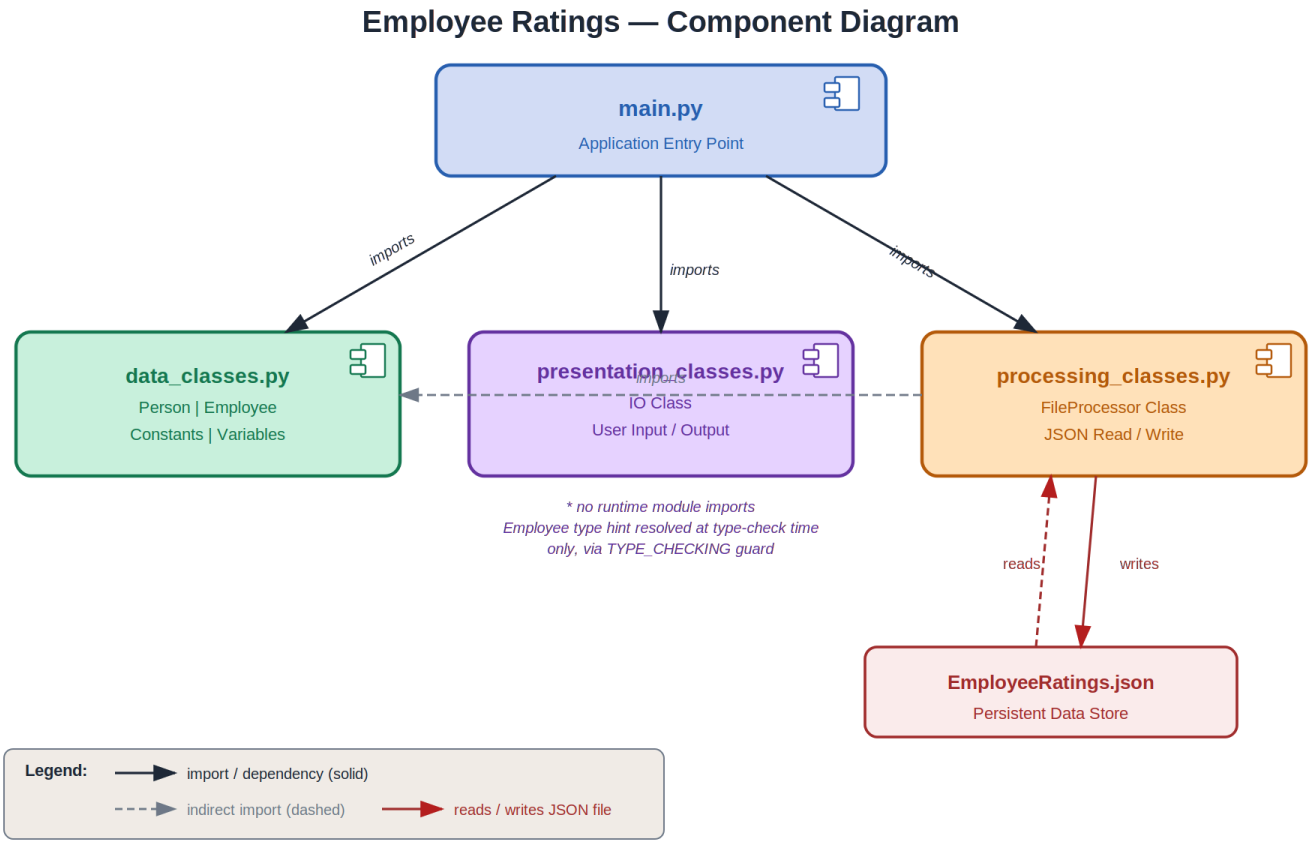


The class diagram shows four classes and their relationships:

- `Person` is the base class providing `first_name` and `last_name` with validation
- `Employee` inherits from `Person` and adds `review_date` and `review_rating`
- `FileProcessor` uses `Employee` objects when reading from and writing to JSON
- `IO` uses `Employee` objects for display and user input collection

All properties in `Person` and `Employee` use the `@property` decorator with setters that validate input before assignment.

Component Diagram

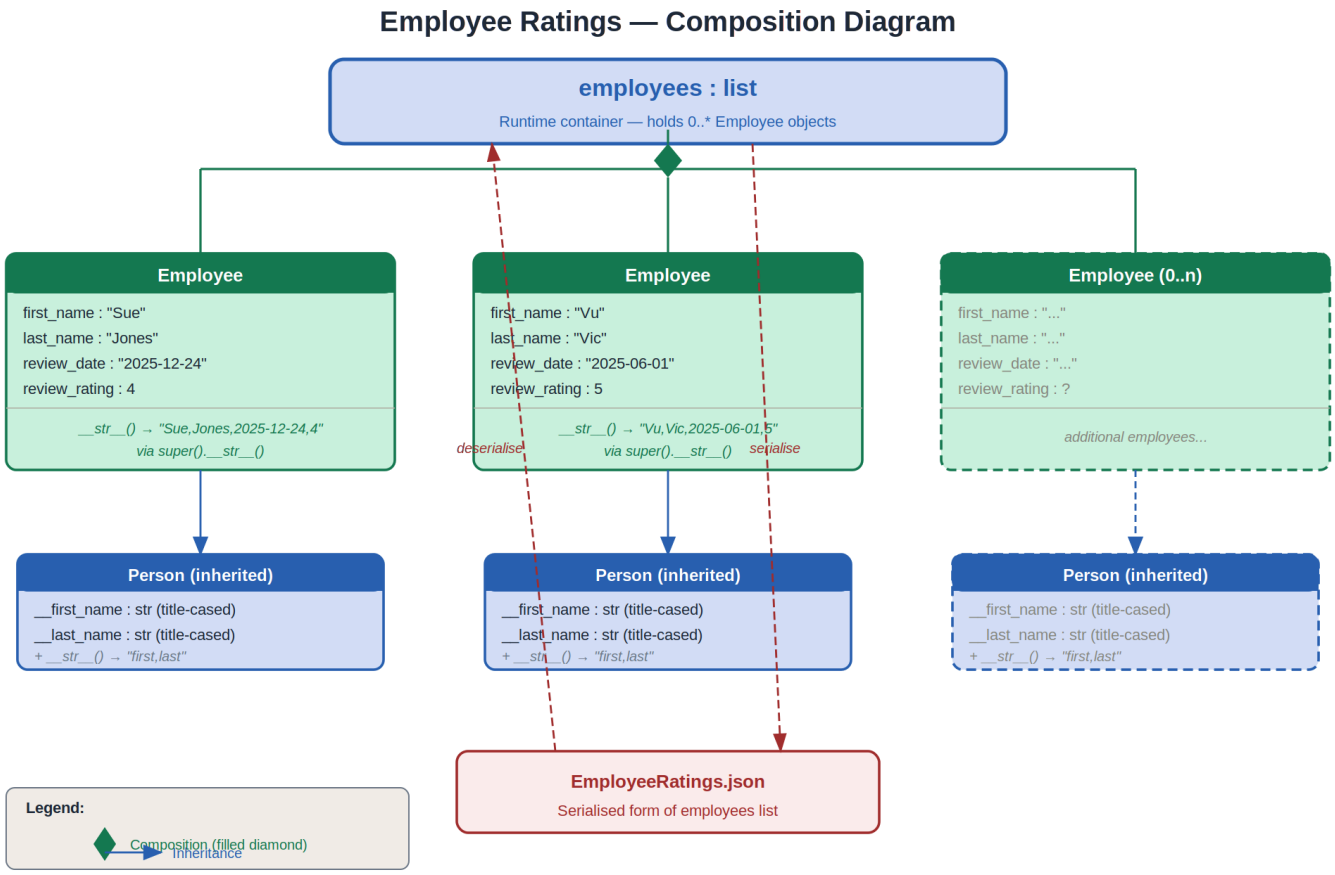


The component diagram shows the module dependencies at runtime:

- `main.py` imports all three supporting modules
- `processing_classes.py` imports `data_classes.py` for type references
- `presentation_classes.py` has no runtime module imports, but Employee is resolved via TYPE_CHECKING
- `processing_classes.py` reads from and writes to `EmployeeRatings.json`

There is no circular dependency between modules.

Composition Diagram



The composition diagram shows the runtime object structure:

- The `employees` list is the central runtime container
- It is composed of zero or more `Employee` objects
- Each `Employee` object inherits its `first_name` and `last_name` fields from `Person`
- The entire `employees` list is serialized to `EmployeeRatings.json` on save and deserialized back into objects on load

Classes and Modules

`data_classes.py`

`Person`

Base class representing a person.

Member	Type	Description
`first_name`	`str`	Property; validated; title-cased
`last_name`	`str`	Property; validated; title-cased
`__str__()`	method	Returns `"first_name,last_name"`

Validation: names must be alphabetic or empty. Numbers and special characters raise `ValueError`.

`Employee(Person)`

Subclass adding review fields.

Member	Type	Default	Description
`review_date`	`str`	`1900-01-01`	ISO format date; validated
`review_rating`	`int`	`3`	Integer 1–5; validated
`__str__()`	method	–	Calls `super().__str__()` then appends review fields CSV

`__str__()` delegates to `Person.__str__()` via `super()` to produce the full comma-separated output, avoiding field repetition.

`processing_classes.py`

`FileProcessor`

Method	Description
`read_employee_data_from_file`	Reads JSON file; returns list of Employee objects
`write_employee_data_to_file`	Writes list of Employee objects to JSON file

`presentation_classes.py`

`IO`

Method	Description
`output_error_messages`	Displays user-friendly + optional technical error
`output_menu`	Prints the menu string
`input_menu_choice`	`while True` loop; re-prompts on invalid input until 1–4 is entered

Method	Description
`output_employee_data`	Displays all employees with name, date, rating number and label; uses a `rating_descriptions` dict
`input_employee_data`	Collects and validates a new employee record

Data File

`EmployeeRatings.json` stores employee records as a JSON array:

```
[
  {
    "FirstName": "Sue",
    "LastName": "Jones",
    "ReviewDate": "2025-12-24",
    "ReviewRating": 4
  }
]
```

The file must exist before running the application. At minimum it should contain an empty array `[]` or one or more valid records.

Error Handling

Scenario	Exception Raised	Handler
JSON file not found	`FileNotFoundError`	`output_error_messages()`
Malformed JSON file	`json.JSONDecodeError`	`output_error_messages()`
Invalid JSON content	`TypeError`	`output_error_messages()`
File write permission denied	`PermissionError`	`output_error_messages()`
Invalid first or last name	`ValueError`	`output_error_messages()`
Invalid review date format	`ValueError`	`output_error_messages()`
Invalid review rating (not 1-5)	`ValueError`	`output_error_messages()`

The processing layer (FileProcessor) re-raises exceptions upward. The presentation layer (IO) catches and displays them via output_error_messages(). main.py catches exceptions from FileProcessor and routes them to output_error_messages().

Running the Application

Prerequisites:

- Python 3.7 or later
- EmployeeRatings.json must exist in the same directory as main.py

From the terminal, navigate to the project folder and run:

```
python main.py
```

The menu will appear:

```
---- Employee Ratings ------------------------
Select from the following menu:
    1. Show current employee rating data.
    2. Enter new employee rating data.
    3. Save data to a file.
    4. Exit the program.
----------------------------------------------
```

Menu Options:

1. Displays all employee ratings currently in memory
2. Prompts for first name, last name, review date (YYYY-MM-DD), rating (1-5)
3. Saves all records to EmployeeRatings.json
4. Exits the program

Running the Tests

Run all tests at once (verbose):

```
python -m unittest discover -v
```

Or run individual test files:

```
python -m unittest test_data_classes
python -m unittest test_processing_classes
python -m unittest test_presentation_classes
```

Test Coverage Summary

Test File	Class Tested	Tests
`test_data_classes.py`	Person, Employee	24
`test_processing_classes.py`	FileProcessor	10
`test_presentation_classes.py`	IO	21
Total		**55**

Note: Some tests were generated with AI assistance and reviewed manually.